Design review · status after V7.1

Taif / Palm House · the review of 30 Oct 2026 (`docs/review/design_review.typ` → `output/Design_Review_V6.pdf`, tag `review-1`, written against V6.2) · status on 31 Oct 2026 after V6.3 (quick fixes), V7.0 (courtyard enclosure, maid room, roof access and attic, design language) and V7.1 (arched enclosure windows, courtyard arcade) · every critical and major finding: 30 in all, **resolved** 8 · **partly** 7 · **open** 15

Id	Sev.	Where	Status	Version	What was done · what is still open
Y-01	critical	Courtyard · the envelope	**resolved**	V7.0	Enclosure option A: W1 on the gallery edge (on the B3 upstand, +3.82 … +6.75), W2 on the family living's edge, W3 over the living's courtyard glass (+3.32 … +6.75), black steel divided lites; the projection 1.55 m, B3 / R11 / C21–C23 at x 5.80; the courtyard stays open to the sky (`studies/courtyard_enclosure.typ`). V7.1: semicircular heads on all of them (W1 six, W2 two, W3 one) over the three-arch courtyard arcade.
A-01	critical	Ageing · the lift	**resolved**	V6.3	The lift is installed at construction, two stops, sliding landing doors; the shaft stays a placeholder until a supplier's drawing (open item 14, now with the headroom rule ≤ 3.40 m / ≤ 3.70 m with a raised head).
S-03	critical	Site · north and lane setbacks	**partly**	V6.3	The house moved 0.30 m south: 0.20 m to the lane setback line (was 0.05), 0.80 m to the north street line; provisional until the official survey (open item 1). **Open:** the survey, and the owner's choice if a larger margin is wanted (0.63 m south, or 0.30 south + 0.17 west).
G-01	major	GF · living wall (west)	**resolved**	V6.3	The living's west wall solid and full height with the 1.40 m entrance arch and the arched TV niche 4.40 × 0.70; sofa facing west, eye-to-screen 3.47 m; no arches between the colonnade columns.
G-04	major	GF · maid room	**resolved**	V7.0	The kitchen's north wall moved 0.50 m south to y 7.0 (a non-load-bearing partition; the y 6.5 beam B6 stays): maid room 7.28 m² clear, 2.49 m clear at the bed (≥ 6.5 m² / 2.1 m); laundry 6.2 m² clear; kitchen 24.0 m² clear with two runs; service door, counter and hood moved with the wall.
F-01	major	FF · gallery width	**resolved**	V7.0	The projection 1.55 m: the gallery 1.405 m clear at the window, 1.235 m at the three column sleeves (≥ 1.20).
F-02	major	FF · door widths	**resolved**	V6.3	Every bedroom and bathroom door a 1.00 m opening with a 0.90 m leaf (≥ 0.85 clear); exceptions recorded: the guest WC (0.90) and the maid bath (0.70).
F-06	major	FF · family living	**resolved**	V7.0	W2: the family living's courtyard edge glazed from the floor to +6.75 (two casements); the room is a conditioned room.
F-07	major	FF / roof · roof access	**resolved**	V7.0	The main stair continues to the roof (flights 3 / 4 in the same well, trimmers R16–R18, a stair room 2.20 m clear); the attic 1.8 m set back 1.0 m behind the main cornice; the roof zones reserved (`studies/roof.typ`).
X-01	major	Circulation · serving the majlis	**open**		The food route from the east kitchen to the men's majlis still crosses the living (17.5 m, four doors). To decide: an outside service path to a majlis serving door, or accept.
G-02	major	GF · living + dining	**partly**	V7.0	The courtyard well is decided (open sky, glazed edges): the living keeps its courtyard light through W3 (× ≈ 0.64 for the glass) and the clerestories. **Open:** the daylight simulation (open item 15) and the garden door / larger east window (S-04).
G-03	major	GF · kitchen	**open**		The hood's make-up air and the kitchen's exhaust in a positively pressurised house: the MEP engineer (open item 8). The hood now sits at y 10.2–11.2 on the east wall, the extract straight out.
G-07	major	GF · bathrooms	**partly**	V6.3	With the lift at construction the first-floor showers are reachable without stairs. **Open:** a ground-floor shower (the family WC has none).
Y-02	major	Courtyard · size for daily use	**partly**	V7.0	The sky opening is 3.15 × 6.5 m (was 3.35); the courtyard floor 4.5 × 6.3 m clear is unchanged. **Open:** the level play zone ≥ 3 × 3 m beside the sunken seating, the seating for 6–8, the hose point.
Y-03	major	Courtyard · sunken seating safety	**open**		The 0.45 m pit with three open steps and no guard on three edges: to decide (guard, a shallower pit, or a level seating area) with the play zone and the drainage outfall.
T-01	major	Structure · X5	**open**		R6 on two first-floor walls over the men's majlis ribs: the engineer verifies the load path or adds columns.
T-03	major	Structure · soil and foundations	**open**		No soil report: pads, the lift pit, the tower pit and the duct are assumed.
E-01	major	Services · the malqaf's value	**open**		With the house sealed round the courtyard the grille serves only the courtyard unless the casements are open; decide: a courtyard feature, the ERV's filtered intake, or drop it (open item 17); the pergola with it (open item 18).

E-02	major	Services · ventilation and dust	**partly**	V7.0	The envelope is now closable (Y-01) so a filtered, positively pressurised system makes sense; the roof plant zone (14.6 m²) and the ERV position are reserved. **Open:** the MEP design (open item 8).
E-03	major	Services · plumbing	**open**		The plumbing study (open item 9) and the drainage outfall (where the roof, the courtyard and the suite court drain to) are still to do.
E-04	major	Services · solar PV provision	**partly**	V7.0	PV zones reserved on the roof: 24.5 m² on the south wing (y ≥ 22.0, clear of the clerestories' sun) and 20.5 m² on the north block; fixing points before waterproofing; the roof live load noted. **Open:** the inverter / battery room, the conduits, the structural allowance.
E-05	major	Services · generator and essentials circuit	**open**		Generator position, changeover switch and the essentials circuit: the site and roof services plan.
E-06	major	Services · security and the safe room	**open**		Cameras, alarm, intercoms and a safe room: the site plan round.
E-07	major	Services · acoustics of the living	**open**		The double-height living's hard surfaces: finishes and absorption to decide at the detail round (the glazed W1 / W3 add hard surfaces).
E-08	major	Services · fire and life safety	**open**		The single stair opens onto the riwaq and the gallery onto the living void; no alarms, no sprinklers; the maid's window sill 1.50 m. **Next, critical:** interconnected alarms, a smoke vent at the top of the void, the escape strategy, sprinklers to decide; the stair room door to the roof is now a second way out upwards.
S-01	major	Site · parking	**open**		Two shaded spaces inside the plot with a covered walk to the family door: the site plan round.
S-02	major	Site · guest approach	**open**		The guest path from the gate to the foyer: the site plan round.
S-04	major	Site · garden access	**open**		A garden door from the living / dining and a larger east window: with the site plan.
S-05	major	Site · east strip, service yard	**open**		Bins, gas, the service door's yard and the outdoor kitchen path: with the site plan.
S-08	major	Site · tanks, septic, plant	**partly**	V7.0	On the roof: the plant zone, the tank zone (3.3 m², if a gravity reserve is wanted) and the PV reserved. **Open:** the ground tank, septic, the generator and the inverter room on the site plan.

The next open items, in this order

1. **Fire and escape** (E-08, critical): interconnected alarms, a smoke vent at the top of the double-height void, the escape strategy from the first floor (the single stair opens onto the riwaq; the stair room door to the roof is a way out upwards), sprinklers to decide, the maid's window sill.
2. **Courtyard: the sunken seating, the level play zone and the drainage outfall** (Y-03, Y-02, Y-04, E-03): a guard or a shallower pit, a level play zone ≥ 3 × 3 m, the hose point, where the courtyard, the roof and the suite court drain to.
3. **The malqaf's value in the sealed house and the pergola** (E-01; open items 17 and 18): a feature only, the ERV's filtered intake, or drop it; the pergola kept, lightened or dropped.
4. **Gym: the treadmill clearances and vibration** (F-09, X6, X5): the arm's clearances and the majlis roof's vibration with the engineer; rubber floor over the whole gym.
5. **Bedroom 1's bathroom route** (F-05, open item 16): 6.9 m from the family bath; accept, or review in the detail round.
6. **The food route from the kitchen to the men's majlis** (X-01): an outside service path to a majlis serving door, or accept the route through the living.
7. **The site and roof services plan** (S-01 … S-08, E-04 … E-06): parking, the guest approach, the garden door, the service yard, the outdoor kitchen path, tanks, septic, generator, inverter room, security, the safe room; the roof zones of `studies/roof.typ` tie into it.
8. **Structure X5 and the soil report** (T-01, T-03, engineers): R6's load path, the pads, the pits.
9. **The legal survey** (S-03, open item 1): the setback margins (0.20 m to the lane, provisional) confirmed before the wall faces are fixed.

Minor findings, suggestions and strengths are unchanged in the review; of the minors, T-02 (X3) is resolved (V6.3) and A-02 (thresholds) is unchanged. Nothing here is for construction: concept design, all sizes assumed until the engineers confirm.